

Mapping E-R Model to Relational Model

Comprehensive Notes with Visuals

1 Basic Mapping Rules

- Each **entity set** and **relationship set** has a unique relation schema.
- Attributes become columns in the schema, each with a unique name.
- Both E-R and relational models are logical abstractions and can be converted between each other.

2 Representing Entity Sets

Strong Entity Sets

A strong entity set becomes a schema with the same attributes.

Example

```
course(course_id, title, credits)
```

Weak Entity Sets

A weak entity set includes a foreign key for its identifying strong entity set.

Example

```
section(course_id, sec_id, sem, year)
```

3 Representing Relationship Sets

Many-to-Many

Represented as a schema with:

- Primary keys of participating entity sets.
- Descriptive attributes of the relationship.

Example

```
advisor(s_id, i_id)
```

Many-to-One / One-to-Many

- If total participation on the “many” side, add a foreign key attribute to the many side.

Example

Instead of creating `inst_dept`, add `dept_name` to `instructor`.

One-to-One

- Add foreign key to either side.
- If partial participation on the “many” side, using foreign key may cause NULLs.

4 Combination of Schemas

1. **One-to-One**: Combine into AR, B or A, BR.
2. **One-to-Many**: Combine into A, BR.
3. **Many-to-One**: Combine into AR, B.
4. **Many-to-Many**: Keep all three schemas A, B, R.

Note: Works best with *total participation*. Partial participation may cause NULLs.

5 Composite and Multivalued Attributes

Composite Attributes

Flatten into separate attributes.

Example

```
instructor(ID, first_name, middle_initial, last_name, street_number,  
street_name, city, state, zip_code, date_of_birth)
```

Multivalued Attributes

Represent as a separate schema with:

- Primary key of the original entity.
- The multivalued attribute.

Example

```
inst_phone(ID, phone_number)  
Tuple examples: (22222, 456-7890), (22222, 123-4567)
```

Special Case Optimization

If an entity has only one attribute other than the PK and that attribute is multivalued, keep only the multivalued relation.

Example

```
time_slot(time_slot_id, day, start_time, end_time)
```